

Global Support — Interview Assignment

Email Bounce Rate Investigation Report

1. Investigation Overview

A client reported an increased email bounce rate during May 2018. This investigation analyzes SMTP response data across 6 months (February to August 2018) to identify the root cause and propose a solution.

2. Investigation Methodology

The investigation followed a structured approach:

- Validate the reported increase in bounce events.
- Compare SMTP response trends across all months.
- Identify the SMTP error code contributing most to the increase.
- Analyze affected email domains.
- Develop a root cause hypothesis supported by the available data.

3. Investigation Steps

Step 1 — Understanding The Data

- The dataset contains 167 rows of SMTP response data across 6 months
- Column 1 = UNIX timestamp converted to readable dates
- Remaining columns = SMTP response codes per email domain
- Key domains include gmail.com, hotmail.com, centrum.sk, seznam.cz, yahoo.com and others
- SMTP code 200 = successful delivery, all other codes = bounce or error

Step 2 — Converting Timestamps and Grouping By Month

- UNIX timestamps were converted into readable dates before analysis.
- Data was grouped by month to compare bounce rates across the 6 month period
- Total successful deliveries and total bounces were calculated per month
- Bounce rate percentage was calculated as total bounces divided by total emails times 100

Step 3 — Comparing Monthly Bounce Trends

Monthly SMTP bounce events were compared across the six-month period.

The analysis showed that May 2018 experienced a significant increase in bounce events compared with previous months, confirming the client's observation. Additional elevated bounce activity continued into June before gradually decreasing in July and August.

This comparison established May as the starting point for detailed investigation into SMTP response codes and affected email domains.

Step 4 — Identifying Root Cause By Error Code

Bounce codes in May 2018 were analyzed in detail:

- Code 421 — Server temporarily unavailable: 204 occurrences
- Code 451 — Server error try again: 210 occurrences

- Code 452 — Insufficient storage gmail.com: 721 occurrences
- Code 499 — Connection timeout: 266 occurrences
- Code 550 — Invalid email address hard bounce: 636 occurrences
- Code 552 — Mailbox full: 450 occurrences
- **Code 554 — Transaction failed: 2,835 occurrences ⚠ DOMINANT ERROR**
- Code 605 — Unknown delivery error: 639 occurrences

Code 554 accounts for the largest share of bounces in May 2018 at 2,835 occurrences — significantly higher than any other error code.

Step 5 — Identifying Root Cause By Domain

Breaking down error code 554 by email domain revealed a critical finding:

- centrum.sk — 1,884 occurrences of error 554 in May 2018
- Other domains — 884 occurrences combined
- centrum.sk contributed 66% of all 554 errors in May

Comparing centrum.sk 554 errors across months:

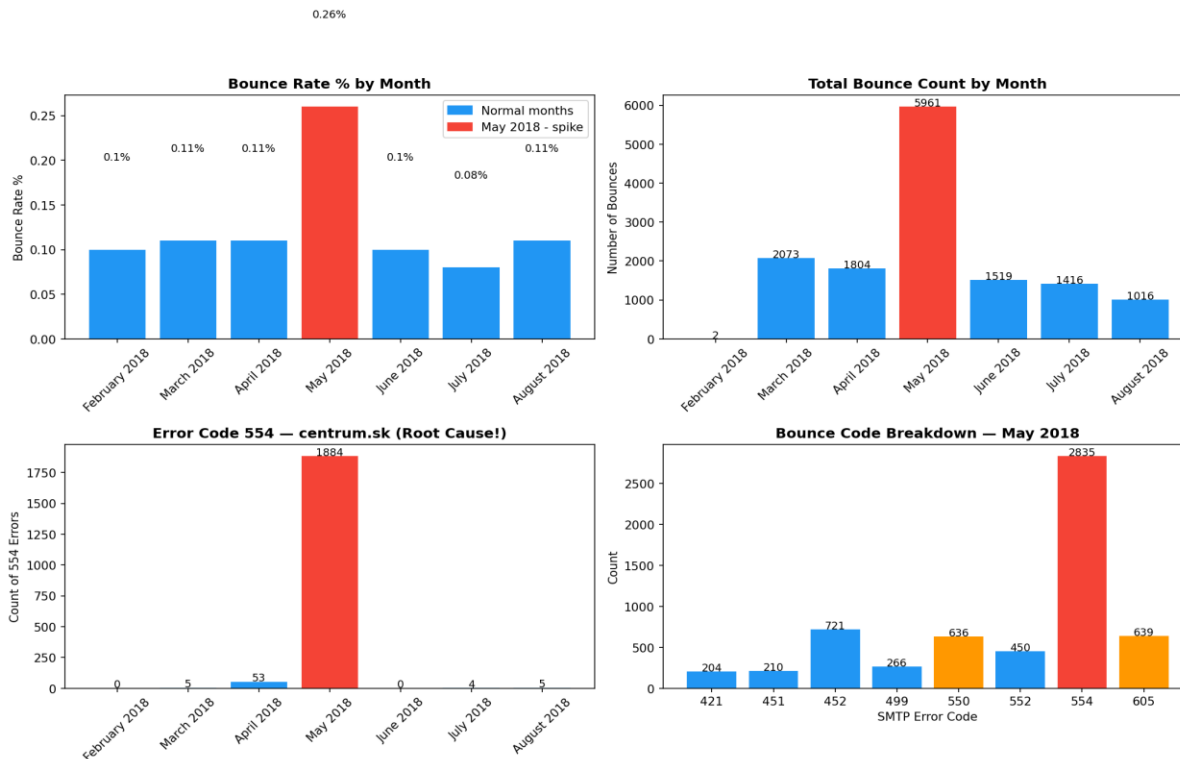
- February 2018: 0
- March 2018: 5
- April 2018: 53
- **May 2018: 1,884 — massive spike**
- June 2018: 0
- July 2018: 4

The spike is isolated to May 2018 and almost entirely concentrated on centrum.sk domain. This is not a gradual trend but a sudden sharp increase that reversed in June.

4. Data Visualization

The following charts summarize the key findings from the investigation:

Email Bounce Rate Analysis — Global Support Investigation



Charts show: Monthly bounce trend confirming May spike, SMTP error code analysis showing code 554 as dominant error, and domain analysis confirming centrum.sk as the primary affected domain.

5. Root Cause Analysis

Based on the data analysis the most likely root cause is:

centrum.sk email domain experienced a server-level policy change or technical configuration update in May 2018 that caused it to reject email deliveries with SMTP error 554 (Transaction Failed). This is supported by the following evidence:

- Error 554 on centrum.sk spiked from 53 in April to 1,884 in May — a 35x increase
- The spike is isolated only to centrum.sk and only to May 2018
- The error completely disappeared in June suggesting a temporary configuration issue
- Error 554 indicates that the receiving server rejected the email during the SMTP transaction. Additional investigation is required to determine whether the rejection was caused by recipient server policies, sender reputation, authentication, or message content.
- Other domains show normal bounce rates in May confirming the issue is domain specific

Secondary factors contributing to the overall bounce spike in May:

- Code 452 on gmail.com increased to 721 suggesting storage or rate limiting issues
- Code 550 increased to 636 suggesting some invalid email addresses in the list
- Code 605 increased to 639 suggesting unknown delivery errors on multiple domains

6. Recommended Next Steps

Immediate Actions

- Contact centrum.sk postmaster or email administrator to understand why 554 errors spiked in May
- Check if the sending domain or IP address was temporarily blacklisted by centrum.sk in May
- Review email sending practices — frequency, volume and content — to confirm compliance with centrum.sk policies
- Verify if centrum.sk made any server configuration changes in May 2018

Email List Hygiene

- Clean the email list to remove invalid addresses causing 550 hard bounce errors
- Implement email validation before adding new addresses to the mailing list
- Remove or separate centrum.sk addresses and test delivery before resuming full campaigns

Monitoring and Prevention

- Set up real-time SMTP error monitoring to catch spikes early before they escalate
- Create alerts when bounce rate exceeds 2% for any domain so investigation can begin immediately
- Implement domain-level bounce rate tracking as a standard practice going forward
- Consider using an email deliverability tool like Mailgun or SendGrid which provides automated bounce management

For gmail.com 452 Errors

- Review sending volume to gmail.com — 452 errors suggest rate limiting or storage issues
- Implement gradual sending rather than bulk sending to reduce rate limiting
- Consider Google Postmaster Tools to monitor domain reputation with Gmail

7. Assumptions

- The dataset accurately represents SMTP response activity.
- Campaign volume or total email sent data was not available.
- Findings are based solely on SMTP response trends contained in the dataset.

8. Summary

The investigation identified a significant increase in SMTP bounce events during May 2018. Analysis of SMTP response codes showed that error 554 (Transaction Failed) was the dominant contributor to the increase. Further investigation by email domain revealed that the majority of these errors originated from the centrum.sk domain, where error 554 increased dramatically during May before returning to normal levels in June.

Based on the available SMTP response data, the most likely explanation is a temporary recipient-side policy or configuration issue affecting email acceptance at the centrum.sk domain. Recommended next steps include contacting the domain administrator, reviewing sender reputation and authentication records, validating mailing lists, and implementing domain-level SMTP monitoring to detect similar issues earlier.

— End of Report —